

Unit 3 ReviewExponents and Scientific Notation **ANSWER KEY**

Name

Date

Period

Show your work. Circle your final answer.

HOW TO USE THIS. This review covers every learning target from the unit — more than the assessment does. Work it in parts. If you miss two or more in a part, that part is where to spend your study time. The reference sheet has everything you need for each part.

PART A. Product, Quotient and Power Laws

Lessons 1–3 · MA.7.NSO.1.1 · 8 questions

1. Write as a single power: $6^4 \cdot 6^5$

$$6^9$$

Like bases: add the exponents. The base stays 6.

2. Write as a single power: $\frac{3^8}{3^2}$

$$3^6$$

3. Write as a single power: $(5^2)^4$

$$5^8$$

Power of a power: multiply the exponents.

4. Find the value: $\left(\frac{3}{4}\right)^3$

$$\frac{27}{64}$$

Power of a quotient: raise the top and the bottom.

5. Find the value: $(2 \cdot 5)^3$ and $2 \cdot 5^3$. Are they equal?

$$(2 \cdot 5)^3 = 10^3 = 1,000 ; \quad 2 \cdot 5^3 = 2 \cdot 125 = 250 . \text{ No.}$$

Without parentheses only the 5 is cubed.

6. Find the value: $9^0 + 4^0$

$$2$$

Each zero power is 1, so $1 + 1 = 2$ — not 0 and not 13.

7. Find the value of n : $\frac{7^n}{7^2} = 7^3$

$$n = 5$$

$$n - 2 = 3.$$

8. Which expression is equivalent to $4^3 \cdot 4^2$?

- ☒ A. 4^5
☐ B. 16^5
☐ C. 4^6
☐ D. 16^6

A

PART B. Zero and Negative Exponents

Lessons 4–5 · MA.8.NSO.1.3 · 7 questions

9. Rewrite with a positive exponent: 7^{-3}

$$\frac{1}{7^3}$$

10. Rewrite with a positive exponent: $\frac{1}{(-5)^{-4}}$

$$(-5)^4$$

The power moves up; the base keeps its parentheses.

11. Find the value: 3^{-4}

$$\frac{1}{81}$$

Reciprocal of 3^4 . Not -81 and not -12 .

12. Find the value: $(-4)^{-3}$

$$-\frac{1}{64}$$

1 over $(-4)^3 = 1$ over -64 . The sign comes from the base.

13. Find the value: $\left(\frac{2}{5}\right)^{-2}$

$$\frac{25}{4}$$

Flip the fraction, then square it.

14. Find the value: $\frac{6^0}{6^{-2}}$

$$36$$

$0 - (-2) = 2$, so 6^2 . Or: 1 over 6^{-2} is 6^2 .

15. Find the value of n : $2^n = \frac{1}{64}$

$$n = -6$$

16. Find the value: $(2^3)^2 \cdot 2^{-4}$

4

$$2^6 \cdot 2^{-4} = 2^2.$$

17. Find the value: $\frac{5^{-2} \cdot 5^6}{5^3}$

5

$$-2 + 6 - 3 = 1.$$

18. **SSDD.** Every part below is about the expression $4^{-2} \cdot 4^5$. Same surface, four different questions: rewrite, evaluate, judge an equivalence, solve for an exponent. Only part c has a yes/no answer and only part d has a negative one.

- a. Write it as a single power.

4^3

$$-2 + 5 = 3.$$

- b. Find its value.

64

- c. Is it equivalent to $\frac{4^5}{4^2}$? Explain.

Yes — 4^{-2} is $\frac{1}{4^2}$, so the product is $\frac{4^5}{4^2} = 4^3$ either way.

- d. Find n so that $4^{-2} \cdot 4^n = 1$.

$n = 2$

$$-2 + n = 0.$$

19. Select ALL of the expressions equivalent to 3^{-2} .

☐ A. -6

☐ B. -9

☒ C. $3^2 \cdot 3^{-4}$

☒ D. $\frac{3^4}{3^6}$

☒ E. $\frac{1}{9}$

☒ F. $\left(\frac{1}{3}\right)^2$

C, D, E and F

20. Aryella says that 10^0 and 134^0 are equal. Is she right? Explain.

Yes. Any nonzero base to the zero power is 1, so both equal 1.

21. Write as a single monomial: $x^7 \cdot x^5$

$$x^{12}$$

22. Write as a single monomial: $(2m^4)^3$

$$8m^{12}$$

The 2 is cubed too.

23. Write as a single monomial: $(3a^2b)(5ab^4)$

$$15a^3b^5$$

24. Write as a single monomial: $\frac{18x^6y^3}{6x^2y^3}$

$$3x^4$$

y^3 over y^3 is $y^0 = 1$.

25. Rewrite with positive exponents only: $4x^{-3}$

$$\frac{4}{x^3}$$

Only the x moves.

26. Rewrite with positive exponents only: $(2y^{-2})^3$

$$\frac{8}{y^6}$$

27. Rewrite with positive exponents only: $\frac{a^3}{a^8}$

$$\frac{1}{a^5}$$

$$3 - 8 = -5.$$

28. Which expression is equivalent to $(5x^2)^3$?

- ☐ A. $5x^6$
☒ B. $125x^6$
☐ C. $125x^5$
☐ D. $15x^6$

B

PART E. Large Numbers in Scientific Notation

Lesson 8 · MA.8.NSO.1.4 · 5 questions

29. Write 7,400,000 in scientific notation.

$$7.4 \times 10^6$$

30. Write 205,000,000,000 in scientific notation.

$$2.05 \times 10^{11}$$

The interior zero stays in the coefficient.

31. Write 8.3×10^5 in standard form.

$$830,000$$

32. How many times larger is 9×10^7 than 3×10^4 ?

$$3 \times 10^3 = 3,000 \text{ times}$$

$$9 \div 3 = 3 \text{ and } 10^7 \div 10^4 = 10^3.$$

33. Dana writes 45,000 as 45×10^3 . Is her expression equal to 45,000? Is it scientific notation? Explain.

Equal, yes. Scientific notation, no — 45 is not less than 10. The correct form is 4.5×10^4 .

PART F. Small Numbers, and How Many Times

Lesson 9 · MA.8.NSO.1.4 · 6 questions

34. Write 10^{-5} as a decimal.

$$0.00001$$

A 1 in the fifth decimal place.

35. Write 0.00093 in scientific notation.

$$9.3 \times 10^{-4}$$

The 9 is in the fourth decimal place.

36. Write 6.1×10^{-7} in standard form.

$$0.00000061$$

37. Which is greater, 4.8×10^{-3} or 9.1×10^{-4} ? Explain.

$$4.8 \times 10^{-3} \text{ — the exponent } -3 \text{ is greater than } -4; 0.0048 > 0.00091.$$

38. How many times larger is 8.4×10^{-2} than 2.1×10^{-6} ?

$$4 \times 10^4 = 40,000 \text{ times}$$

$$8.4 \div 2.1 = 4 \text{ and } -2 - (-6) = 4.$$

- 39.** A computer displays a number as $2.7\text{E}-5$. Write it in scientific notation and in standard form.

$$2.7 \times 10^{-5} = 0.000027$$